

The Social Construction of Generative AI Prompts

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ABSTRACT

As text-to-image AI tools grow in capability and widespread use, research has focused on studying individualistic user prompt crafting strategies. Recognizing that technologies are socially constructed, this paper examines prompt engineering through a social lens. We propose reframing prompt engineering as a socio-cultural practice shaped by collective knowledge building. Through qualitative analysis of 19 semi-structured interviews with members of the MidJourney community, a text-to-image generative AI tool, we identify four socio-engagement themes: proprietary/solitary, derivative, collaborative, and provocative prompting. These themes reveal a space of social engagement modes based on personal values and motivations from individual exploration to influencing the prompt community and highlight a fine line between being inspired by others' prompts and maintaining creative ownership. We argue that understanding distinct social engagement preferences can inform the design of AI tools to facilitate transparent prompt reuse mechanisms, integrate collaborative features, or preserve ethical concerns about prompt sharing.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → **Artificial intelligence**; • **Applied computing** → **Arts and humanities**.

KEYWORDS

Prompt Engineering, Text-to-Image, Communities of Practice, Generative AI

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1 INTRODUCTION

Recent advancements in text-to-image AI tools [27, 29, 39], which utilize user-generated text inputs referred to as prompts, has led to a swell of research focusing on the composition, complexity, and effective strategies of prompting [23, 25, 31, 45]. Such research, while yielding critical insights, often studies prompting as an isolated and

individual experience. However, the emergence of social groups and online communities, such as Facebook Groups and Discord servers dedicated to discussing and sharing prompts and prompt documentation [1, 2, 14, 38], indicates a social dimension underlying prompt engineering.

Drawing from the Social Construction of Technology, we argue that prompts are socially constructed and shaped by the interests and values of diverse groups [13], rather than happening in a social vacuum. Motivated by this perspective, this paper explores the broader socio-cultural context of prompting - focusing particularly on the social dynamics present within MidJourney, a text-to-image generative AI tool. We seek to answer: How do users engage in the practices of prompt adaptation, sharing, and peer learning within text-to-image AI communities?

Our study presents findings from a qualitative analysis of semi-structured interviews with 19 existing members of the MidJourney community, a popular platform for text-to-image generation. Our contributions are as follows: First, our findings suggest that prompting is not merely an individual activity but a socially constructed and shared practice, extending beyond ideal prompt-writing strategies to encompass social dimensions and collective knowledge building. Second, we characterize different orientations of engagement through which community members learn from and contribute to the community. We uncover four themes of social engagement: *proprietary/solitary prompting*, *derivative prompting*, *collaborative prompting*, and *provocative prompting*. Lastly, we discuss how reframing prompt engineering as a socially situated practice surfaces opportunities for designing AI tools that foster learning and creativity, facilitate transparent prompt reuse mechanisms, integrate collaborative features, or preserve ethical concerns about prompt sharing.

2 RELATED WORK

The concept of "prompt engineering" has gained traction in research, initially emerging in natural language processing for programming tasks [34] and evolving to represent the nuanced process of crafting the right prompt to yield specific outputs from large language models [37]. This area of study has predominantly focused on individual aspects of prompt creation, encompassing various challenges such as task granularity, request scoping, and addressing users' misconceptions about prompt history and syntax [18, 19].

To enhance the effectiveness of prompt-based interactions, numerous studies have concentrated on strategies and tools. They have examined factors influencing prompt performance, including the negative impact of lengthy task instructions [28] and the role of word order in model accuracy [24]. Tools designed for individual use, featuring constrained vocabularies, menu-driven interfaces, and block-based programming, have been developed to aid users in

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prompt formulation and writing [22, 33]. Additionally, Zamfirescu-Pereira et al. [45] have examined intuitive prompting behaviors with large language models, focusing on individual user interactions with a conversational agent.

Particularly in text-to-image AI tools, prior research has extensively analyzed the efficacy of different prompting techniques in diffusion models, leading to the development of specific guidelines [12, 23, 44]. Notably, Liu and Chilton put forth comprehensive guidelines addressing elements like prompt phrasing, seeds, iterations, and style [23]. Sanchez's work provided further detailing for the semantic structure of prompts [36].

The integration of text-to-image tools into professional creative domains has also been a subject of research. Ko et al. have examined the opportunities and challenges artists encounter when using text-based inputs [21], and Tholander and Jonsson have investigated the role of generative machine learning in creative design processes [41]. Chang et al. have focused on the art community's use of text-to-image models, highlighting the artistic value of prompt templates [9].

While these studies offer valuable insights into individual prompt crafting strategies, there's a gap in understanding the social dynamics surrounding prompts in text-to-image online communities. Oppenlaender's work provides an ethnographic look at these communities, identifying key roles like innovators, porters, conservators, service providers, and practitioners [30]. Building upon this line of inquiry, our study aims to explore the social aspects of prompting, investigating the motivations and values that drive the sharing and use of prompts in text-to-image AI communities.

3 METHODS

3.1 MidJourney

MidJourney, a text-to-image AI tool, has garnered widespread attention, boasting over 10 million users on its Discord server as of February 2023 [26]. Its popularity and the vibrant user interactions it facilitates make it an ideal subject for studying user engagement with text-to-image AI tools. Hosted on Discord, a platform popular especially in the gaming community, MidJourney's server hosts several active channels where users can generate and view images, with prompts and outputs flowing in near real-time. Users can input text into a command bar, leading to the display of a 2x2 grid of four distinct, real-time refining images based on the prompt. Users can further interact with these images, selecting one for variations or upscaling to achieve a higher-resolution version with more detail.

For this research, we concentrate on MidJourney due to its reliance on textual prompts and its relatively simple interface, with minimal parameter adjustments. While acknowledging the diverse capabilities and interfaces of other models, including those with extensive parameters and detailed image editing features, MidJourney's substantial public usage [3, 20, 35] and the strong community that has formed around it as of August 2022 make it a focal point for our study.

3.2 Participant Recruitment

Our recruitment involved social media outreach and snowball sampling. In July 2022, we reached out to 110 individuals active in a Facebook group dedicated to MidJourney. This group serves as a

platform for members to share their art created with MidJourney, exchange advice, explore new prompts, and pose questions. Each user we contacted had posted at least one artwork in the group in the month before our study. Additionally, we approached 20 Instagram public pages featuring recent "#MidJourney" tags for broader outreach.

This study received institutional IRB approval and was conducted with adult participants residing in the US. The focus on US-based participants was deliberate, given our study's emphasis on English-language prompts. This approach was to minimize variations in language interpretation affecting our results. We recognize the importance of expanding future research to include diverse global perspectives and languages.

We crafted a recruitment message detailing the study's purpose and participant requirements. This message was sent privately to prospective participants, and interviews were scheduled with those who expressed interest. In total, 19 participants were recruited, aged between 30-45 years, including 13 males, 4 females, and 2 non-binary individuals. To verify their active use of MidJourney, participants were asked during the interview to log into their MidJourney accounts and showcase their work. We operated under the assumption that the profiles they logged into and the work displayed were their own. All confirmed participants received a \$20 Amazon gift card as a token of appreciation for their participation.

3.3 Study Design

Participants were asked to fill out a pre-interview questionnaire the day before the interview to provide demographic information. We conducted all semi-structured interviews 1-on-1 and remotely through Microsoft Teams, primarily due to the geographical dispersion of our participants as they were members of an online community. With participants' consent, all sessions were video recorded, including the screen activities. The first author conducted all of the interviews. Our research involved the use of MidJourney, for which access required a basic subscription of \$10 per month at the time of our study.

We conducted semi-structured interviews combined with retrospective walk-throughs. Each session was structured into two parts. In the first part, we asked participants about their backgrounds, experiences with MidJourney, and their goals for using the platform. This initial part of the session did not involve screen sharing and served both as an introduction and a foundation for the subsequent part. In the second part, we asked participants to share their screens and walk us through their previously generated visual outputs on their MidJourney profile. As participants showcased their work, we asked questions about their process for writing prompts, how they evaluated the output images, their strategies for refining prompts, the challenges they faced, and the resources they used. This part was integral for providing context to the participants' responses, as it allowed them to reflect and articulate their processes and decisions in direct relation to the visuals they had created. It also enabled us to observe their work and engage in situational follow-up questions, thus grounding the interview responses with their visual outputs.

3.4 Data Analysis

For our thematic analysis, we employed an inductive approach [42] following Braun and Clarke's guidelines [8]. The total interview

Table 1: Thematic Codebook for the Modes of Social Engagement within the MidJourney Community

Theme	Description	Example Quotes
Proprietary/Solitary Prompting	This orientation is characterized by expressing a sense of ownership over one’s prompts and a reluctance to use or replicate the work of others. It reflects a preference for self-exploration to maintain originality and control over one’s creative outputs, often leading to a more solitary approach to prompt creation and usage.	<i>“I kind of feel really uncomfortable with doing variants from other people’s work.” P3</i>
Derivative Prompting	This orientation is characterized by treating others’ work and prompts as examples that can be adapted to meet personal creative needs. Individuals with this orientation comfortably use existing prompts as a foundation, modifying them to align with their unique ideas and stylistic preferences.	<i>“I simply look at things that I like that people do and then use that prompt with my own content.” P16</i>
Collaborative Prompting	This orientation prioritizes collaboration and collective learning over individual originality. Participants who exhibit this orientation express interest in the process of sharing, adapting, and building upon each other’s ideas and sense of belonging to a community.	<i>“We’re constantly bouncing off each other’s prompts and sharing tips and tricks to try and just truly explore as far as we can together.” P6</i>
Provocative Prompting	This orientation is characterized by the strategic presentation of one’s artwork or prompts, this behavior aims to influence or spark new ideas and trends within the community. Those exhibiting this orientation may not engage in direct collaborations, instead focusing on inspiring others through their individual contributions and provocations.	<i>“So I’m assuming I have a bit of influence on that side of things through my art I’ve noticed and I strategically do some of my prompts in the main chat when I want other users to encourage that.” P2</i>

time spanned 17 hours, with an average duration of 55.11 minutes (std = 9.68). The initial coding was led by the first author and was refined through collaborative discussions with the second author, ensuring that emerging themes accurately reflected our observations. This process then involved all three members of the research team in iterative discussions, aiming for consensus on codes while addressing potential biases from early-stage coding. Disagreements were resolved by revisiting and redefining code definitions and considering additional data points. After coding, the first author organized the refined codes into broader themes, further refined through team discussion until unanimous agreement was reached on their representation of the data. Through this process, We identified four themes of social engagement: (1) Proprietary/Solitary, (2) Derivative, (3) Collaborative, and (4) Provocative (see Table 1).

Our analysis revealed additional dimensions around individualistic prompt structures and refinement processes as well as the social dimensions [25]. However, given this paper’s focus on prompt engineering as a socially constructed practice, we concentrate our reporting solely on the findings pertaining to social engagement.

4 FINDINGS

In our analysis of the interview data, four distinct themes emerged regarding community engagement, sharing, and usage of prompts: *Proprietary/Solitary*, *derivative*, *collaborative*, and *provocative prompting*. These themes are defined by varying degrees of willingness or openness to share or use prompts as demonstrated by the two dimensions: Prompt Sharing and Prompt Reuse, depicted in

Figure 1. We note that these themes are not permanent characteristics of individuals but rather temporary, context-dependent orientations reflecting preferences in collaboration with peers, sharing, or using prompts and artwork. These identified themes shed light on the primary modes of exchange, reuse, and participation that users adopt in the dynamic environment of communal knowledge-sharing, particularly in relation to MidJourney.

4.1 Proprietary/Solitary Prompting: Refraining from reusing others’ prompts

Five participants exhibited strong feelings of ownership over their prompts and an intentional resistance to directly re-using the prompts of others. This manifested in several ways. Some participants expressed clear discomfort with directly re-using the exact prompts created by others, with one quoting that they “*feel really uncomfortable with doing variants from other people’s work*” (P3). For these users, directly taking another’s prompt composition feels inappropriate or “*weird*” (P8).

For some participants, there is a tension between learning from the community and feeling that directly appropriating others’ exact wording crosses an ethical line. As P12 notes: “*mostly it’s the modifiers ... cause the subject you know to me [is important.] I’m not just gonna steal or rip off someone’s subject, right?*” There appears to be a delineation between reusing stylistic modifiers from a prompt versus taking someone’s original subject or concept idea without permission. P8 echoes this dilemma, expressing mixed feelings about reusing prompts: “*like I’m gonna steal your [prompt] and I’m gonna make my stuff with it...and it feels kind of odd... It sounds like*

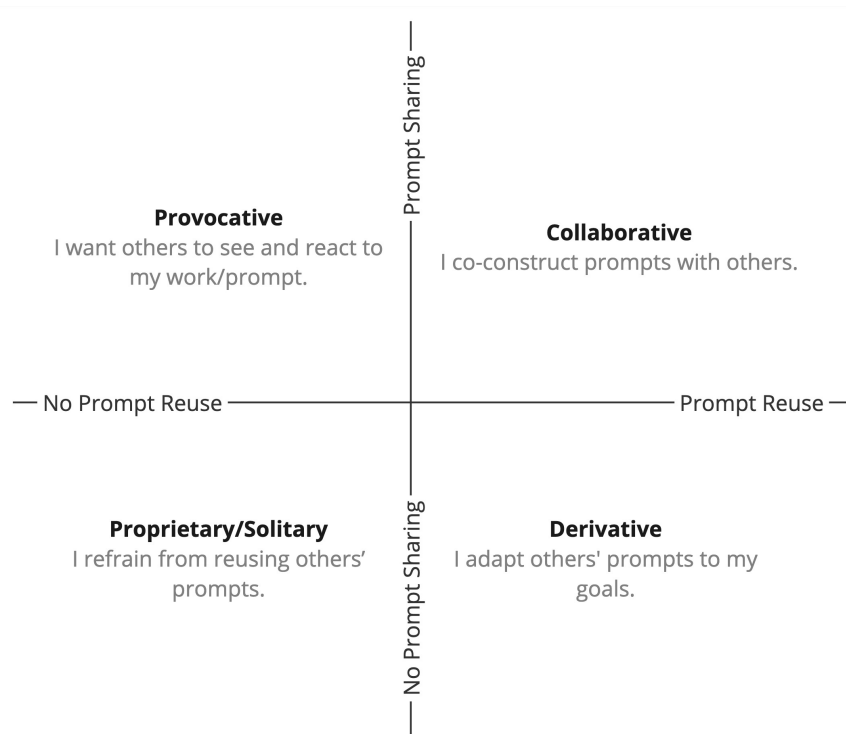


Figure 1: Our space of social engagement, mapping individuals' receptivity to both sharing and reusing prompts

learning. So I guess it's fine." P3 also shares this approach, stating, "I will just look at that [image] and I will appreciate it and I'll look at their prompts. But I would never, like, make a new version."

Initially, some users had strong negative reactions to prompt reuse, feeling other users directly copied their prompts without permission. As P8 described: "Somebody will literally take my prompts, copy it right in front of me and I just say, hey, wait a second..." This echoed P7's feeling of prompts being taken, stating "at first you're like, hey, that's mine." However, for P8 their perspective on these reuse cases shifted over time to acceptance. They explain that when others "add, like, one word and they'll get a better image," it sparked seeing adaptations as mutually beneficial collaboration rather than cheating, noting: "it almost becomes a collaboration because I'm learning from this person and it's not adversarial." Despite this transition in viewpoint, for other users like P7, underlying fears of losing credit persisted. P7 describes intentionally "deleting" elements after sharing prompts, showing concerns about losing control or credit for idea origination.

Finally, some participants (n=3) favored solo trial-and-error over communal resources. As P12 explains, "I like to try to figure stuff out for my own and you know, skim through things and." This indicates that for some, self-driven experimentation is more fulfilling than taking inspiration from others' discoveries or prompts. Similarly, P18 describes purposefully distancing themselves from the community feeds, noting they "have kind of enjoyed it more as like a solitary activity."

4.2 Derivative Prompting: Adapting others' prompts to personal goals

Some participants mentioned initiating their creative process by selecting a prompt that produces a desired visual effect, and then modifying it to suit their personal vision. For example, P10 explains their approach: "Sometimes I'll take a prompt and if I like everything, I'll do nothing but change the subject." Participants find examples shared on social media or Discord channels valuable for inspiration, yet they tend not to engage directly with the social groups themselves. P16 describes this process of adaptation: "They are like a skinless human floating in a circular globe and I could put somebody something else in that 'circular globe' and it would be interesting." Such modifications illustrate how participants transform existing prompts into new creations.

The integration of personal content into existing prompts is a common practice among participants exhibiting this orientation. P12, for example, speaks to this, "I put my own subjects to it to see how it reacts and get to something that I like to look at, not just something that I liked to look at that was theirs." Similarly, P19 focuses on adopting stylistic elements rather than entire concepts: "Mostly the graphic style, because the idea and concept I don't wanna copy this concept from them, but I want to have, for example, the rule of thirds or golden ratio. I saw somewhere on the website I tried to apply that into my work." These statements show a desire to personalize while respecting the original idea.

4.3 Collaborative Prompting: Co-constructing prompts with others

Some participants (n=7) exhibit a culture of peer learning within the community. This culture is exemplified by users like P11, who iterates on existing prompts, as they mentioned that they *"have used other people's prompts and added [their] own detail."* Peer support is also leveraged in overcoming challenges, as P8 described: *"asking friends"* for guidance to resolve prompting difficulties. Such engagement facilitates what P15 referred to as *"practice with engaging community learning,"* highlighting the mutual growth derived from collective interactions.

Integral to these collaborative interactions is the open sharing of ideas, with little focus on ownership, as described by P10: *"We're sharing ideas...I won't get mad at you for using a word I introduced."* This open attitude fosters environments where creativity happens collectively rather than in isolation. Some utilize reciprocal open-source sharing to further implant this culture, like P6 discussing sharing back prototypes to *"enable others to learn from and expand upon"* creations.

Some users direct efforts into organized collective activities by creating and moderating their own Discord servers. For example, P6 described, that their Discord community members have initiated collaborative experiments: *"We have a group of people and we are trying to experiment with pushing image prompts as far as they can go because there's been no defining literature as far as how many references we can have."* Others described ongoing efforts as well, like: *"this is a collaborative effort between me and about a dozen other people that we've been working on for a week."* (P6).

Collaborative prompting is actively utilized as a creative strategy. As P7 described: *"Set a goal and then try and work through the journey of getting there collectively and letting people see like how prompts modify over time."* P9 highlighted *"constantly bouncing off each other's prompts and sharing tips and tricks to try and just truly explore as far as we can together."* This building on existing prompts and blending different individuals' contributions enable what P10 called *"the collaborative element"* of collectively converging ideas into innovative creations within the community.

4.4 Provocative Prompting: Seeking engagement and reactions to works or prompts

Eight participants reported sharing their artwork on platforms like Facebook and Instagram, with motivations ranging from *"put it out there"* (P16), and *"matching the theme of the day"* (P15) to influencing the community's trends (P2, P10, P17). Participants not only sought to introduce innovative techniques and styles but also expressed a sense of fulfillment when these contributions were adopted by the community. P2 articulated this sentiment, stating, *"It's nice to be a part of it [the community]. I appreciate the recognition."*

Several participants took active roles in fostering new styles and ideas by sharing their prompts and techniques. P17, for instance, shared an innovative approach using emojis in prompts, which was well-received by the community. Others, like P2, deliberately presented certain styles or color schemes to guide community trends, stating, *"I have a bit of influence on that side of things through my art...and I do strategically showcase my prompts when I want other*

users to encourage that." Additionally, participants like P10 enjoyed seeing how their ideas were adapted and evolved by others, *"It's great how people are using it cause they'll take the ideas that I've had and they'll turn it into something different."* P4 also shared prompts occasionally, acknowledging the importance of contributing to the community that supports them: *"I try to share some of my prompts every now and again because I feel like I've got to feed back into the ecosystem that's providing for me."*

5 DISCUSSION

In this section, we discuss reframing prompt engineering as a socially situated practice and its implications for fostering creativity and learning, as well as the design of generative AI tools.

5.1 Reframing Prompting as a Socially Situated Practice

The themes emerging from this study collectively support repositioning prompt engineering not merely as a technical skill defined by efficacy standards in an individual's mind, but rather as a socio-technical practice. This perspective aligns with Hutchin's theory of distributed cognition [16], which posits that knowledge extends beyond individual mental models, and with Glaveanu's concept of distributed creativity, emphasizing the dynamic, socio-cultural, and developmental nature of creativity [15]. Additionally, Suchman's work complements this view by demonstrating that learning and using new tools, such as prompt engineering in MidJourney, is fundamentally a social practice [40]. We observed how individual values around and priorities around prompt sharing and reusing shape the practice of prompt creation and adaptation, underlining the social dimension of MidJourney. Such repositioning acknowledges the impact of motivations like community influence, originality, and collaboration on user involvement and echoes findings in research on social media user motivations [6].

We highlight a spectrum of engagement modes, ranging from solitary exploration (e.g., P3) to collaborative prompt crafting (e.g., P9), indicative of diverse interaction styles with the AI systems within the community. This variety reveals emerging norms that delicately balance between drawing inspiration, avoiding direct appropriation, and fostering collaboration. Users inclined toward solo exploration tend to establish distinct boundaries between their own prompts and those of others, preserving a sense of personal ownership over their creative outputs. Conversely, those with a collaborative orientation prioritize shared experiences, favoring communal participation in the development and exploration of prompt creation techniques.

5.2 Fostering Creativity and Learning

Reframing prompt engineering as a socially situated practice highlights the potential for fostering creativity and learning through collaborative interactions within the community. Acknowledging the diverse interaction styles observed, ranging from solitary exploration to collaborative co-creation (e.g., P3 and P9), suggests that platforms and tools can be tailored to accommodate these various learning preferences.

Currently, learning materials for prompt writing primarily support derivative and solo explorations by enabling users to observe

other prompts as evidenced in our findings, or interact with tutorials [1, 2, 14, 38]. While current social media groups (e.g., [4, 5]) facilitate discussions around generative AI, collaborative efforts often occur within smaller, community-built spaces such as Discord servers (e.g., P6 and P10), where users engage in group collaboration. Text-to-image tools, alongside other generative AI tools, have the potential to integrate features that facilitate collaborative prompt writing. Additionally, these tools can incorporate functionalities for content moderation [17] to facilitate user trust [10], within text-to-image tools and generative AI platforms more broadly.

Prompts themselves are often seen as artistic expressions by AI Art practitioners [9]. While community dynamics may influence trends in prompt creation, image generation platforms can empower users to appreciate the diversity in prompt writing styles and desired artwork [25, 36] by providing uniqueness and diversity scores or personalized prompt recommendations. For instance, text-to-image generation tools can encourage solo explorers to experiment with new prompt techniques, inspire derivative prompters to explore novel ideas beyond existing prompts, or facilitate group projects for collaborative prompting. This recognition empowers users to break away from conventional practices while drawing inspiration from others' prompts, offering them an opportunity to explore creative boundaries [11]. The impact of the four distinct orientations on prompts and overall creativity warrants further investigation. Previous research suggests that negative social dynamics can affect group creativity and should be carefully addressed [43]. Therefore, future work should delve deeper into understanding how different interaction styles and motivations influence creativity within online creative communities.

5.3 Designing for a Spectrum of Engagement

Our findings suggest the design of text-to-image AI tools and interventions that support mixed participation modes and offer tailored support based on individual engagement pathways [7]. Clear mechanisms for tracking and crediting contributions can encourage users to transition between orientations. For example, for individuals orienting to solo exploration who might consider engaging in collaborative work, clear and transparent mechanisms for tracking and crediting contributions can provide the necessary confidence and assurance both for using and sharing prompts and encourage social stimulation [32, 43].

Conversely, for users who thrive on collaboration, the AI tool could benefit from incorporating features that enable easy sharing, synchronous and asynchronous collaboration, and feedback. This might include integrated community spaces, and the ability to co-create or modify prompts in real-time with other users. Additionally, for those engaged in derivative and collaborative prompting, features that enhance the discovery of new vocabulary and concepts could encourage these users to extend their range of prompts, promoting variety and further preventing homogeneity of ideas [36].

6 LIMITATIONS AND OPPORTUNITIES FOR FUTURE RESEARCH

While our study presents insights into the modes of participants, it is limited by its focus on a single community of MidJourney users.

Future studies could broaden the scope to include various communities engaged with different generative AI tools. Furthermore, future research could explore how different community engagement types interact and influence creative outcomes. Research could focus on whether a blend of independent experimentation and social peer learning fosters more innovative and diverse creative outcomes. Investigating the evolution of user preferences and practices over time would also provide a richer understanding of the long-term effects of engagement with AI tools on creativity.

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